

Annotation guidelines and resolution protocol (OADS-1K)

Overview.

Annotations assign each URL example to one of six classes used in OADS-1K: **Third-Party Dataset**, **Third-Party Software**, **Author-Provided Dataset**, **Author-Provided Software**, **Project**, and **General URL**. Annotators used the target sentence plus expanded context (preceding and trailing sentences) to classify each URL and followed the rules below.

1. Class definitions

- **Third-Party Dataset:** URLs linking to datasets generated, managed, and distributed by other entities than the authors.
- **Third-Party Software:** URLs linking to software (e.g., tools) developed by people or organizations other than the authors.
- **Author Provided Dataset:** URLs linking to datasets directly generated and provided by the authors.
- **Author Provided Software:** URLs linking to software or tools directly provided by the authors.
- **Project:** URLs linking to a project website containing both data and software or tools, which can be provided by the authors or third parties.
- **General URL:** URLs linking to resources other than a dataset, software/tool, or project.

(Annotators were given the above definitions plus representative examples drawn from Table I.)

2. Primary annotation rules

1. **Context-first:** Labels were assigned using the target sentence and its expanded context (S-1, S0, S+1). Annotators relied on contextual cues, such as phrases like ‘our data are available at...’, ‘we uploaded the code to...’, or ‘the dataset can be downloaded from...’, to determine whether a URL referred to a dataset, software, a combined data-and-software project, or another type of resource.
2. **Author vs. Third Party:** The annotators preferred contextual cues that indicate authorship (phrases like “we made available”, author names linked to repo/contributor lists). If ambiguous, they checked the landing page (see rule 3).
3. **Landing-page verification:** If context alone is ambiguous, annotators followed the URL to inspect its landing page to disambiguate resource type and to check authorship

(unless the link was inaccessible). The annotators recorded the landing-page check in the annotation notes.

4. **Project:** If a URL references a project that hosts both code and data, the annotators labeled it as **Project**.
5. **Single primary label:** The annotators assigned a single label corresponding to the primary role of the URL in the target sentence based on the contextual cue, even when multiple functions are present on the landing page.

3. Annotation workflow and quality control

- Every example was labeled independently by two expert annotators (graduate students) using the rules above. Annotators worked from the same extracted text (target + expanded context) and logged any landing-page checks.
- We computed pairwise agreement; disagreements were flagged for resolution.
- For disagreements, a third expert annotator reviewed the context and landing page. If the third annotator agreed with one of the initial labels, that label was adopted. Remaining conflicts were discussed in a short adjudication meeting; the final label was decided by consensus. Adjudication notes were recorded for edge cases.

4. Special cases / edge rules (quick reference)

- **Author identification:** If context strongly implies author provision → prefer context. If not, check for the authorship on the landing page by matching the paper's author list → label "Author Provided ...," but mark in notes.
- **Missing or dead links:** If the URL was inaccessible and context alone could not disambiguate, label as **General URL** and record in notes.
- **Multiple URLs in one sentence:** (Excluded from OADS-1K; see paper) The annotators annotated only single-URL sentences; multi-URL cases were set aside for future work.